Overview

- We run the submission project. So make sure that it compiles and works as expected. State your assumptions when you solve a problem.
- Please look at the submission section in order to send it in the correct way.
- Be aware that there could be multiple ways to solve a problem.

AI Usage Policy: You may use AI tools while working on this assignment. However, it is essential that you fully understand the code you submit, including its structure and logic. Submissions that are entirely AI-generated without a clear understanding of how they work will not be accepted.

Problem Statement 1

This test is designed to assess your ability to optimize data transfer in a networked game. The objective is to reduce the amount of data sent over the wire, while still maintaining a smooth and responsive experience for the player.

Build a Simple Level:

- Feel free to design the way you prefer based on the attached video. Please do local simulation, no need to connect to any multiplayer server/backend ● In this video, “My Player” is controlled by W, A, S, and D and “Remote Player” just receives optimized positions via a function call.

Video Link

Problem Statement

- Write logic to reduce the amount of data sent to the Remote player. As you can see in the video. In a normal case, we would be sending 3 floats which would be 96 bits.

- On the receiving end (RemotePlayer), write logic to interpret these values and generate the position again to move the player.
- Write debug statements to print the sent position, size of data and received position (unencrypted). [Shown in the console log-in video]
- The movement of the remote player should look smooth.
- **NOTE**: This is simulated locally. No need to add any networking library / Netcode

Submission

- You should upload the Unity project to Google Drive and share the link. Make sure to give access / public links to the project.
- Uploading this to a public repository/portfolio is not allowed.
- Please delete the Library folder before sharing the project.
- Please append your name with the project folder name. For example, `LILA_Unity_Avinash_Pandey`
- Below is folder structure to follow:

```
Assets/
├── Problem_1/
└── Problem_2/
```

Problem Statement 2

You have a famous multiplayer shooter game. You have been assigned to work on the Weapon System of the game. Please write down the class structure for the feature considering architecture, state management and communication between them.

- Player
    - Can equip two primary guns and one secondary gun
    - Can fire the weapon
- Weapon
    - Feel free to make assumptions about the attributes of a shooter weapon (Ammo, magazine size, fire rate, etc).
- UI
    - Hud which shows currently equipped Weapons
    - Shows current & Total ammo of guns

Submission

- Share C# classes written inside the same Unity project as above. No need to do anything on the scene / UI side. Just plain C# classes.

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